EXPERIMENT 6

MATLAB-BASED MODELLING OF FREQUENCY MODULATION (FM) AND PHASE MODULATION (PM)

AIM

To generate and analyse Frequency Modulation (FM) and Phase Modulation (PM) signals using MATLAB, and to plot the message, carrier, FM, and PM waveforms.

SOFTWARE REQUIRED

MATLAB

PROGRAM CODE

```
clc;
clear;
close all;

Am = 1;
Ac = 1;
fm = 5;
fc = 50;
beta = 5;
kf = 5;
kp = 5;
fs = 1000;

t = 0:1/fs:1;

m = Am*cos(2*pi*fm*t);          % Message signal
c = Ac*cos(2*pi*fc*t);          % Carrier signal

fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM signal
pm_sig = Ac*cos(2*pi*fc*t + kp*m);                % PM signal

Df = beta*fm;
fprintf('Frequency Deviation = %d Hz\n', Df);
fprintf('FM Modulation Index = %d\n', beta);

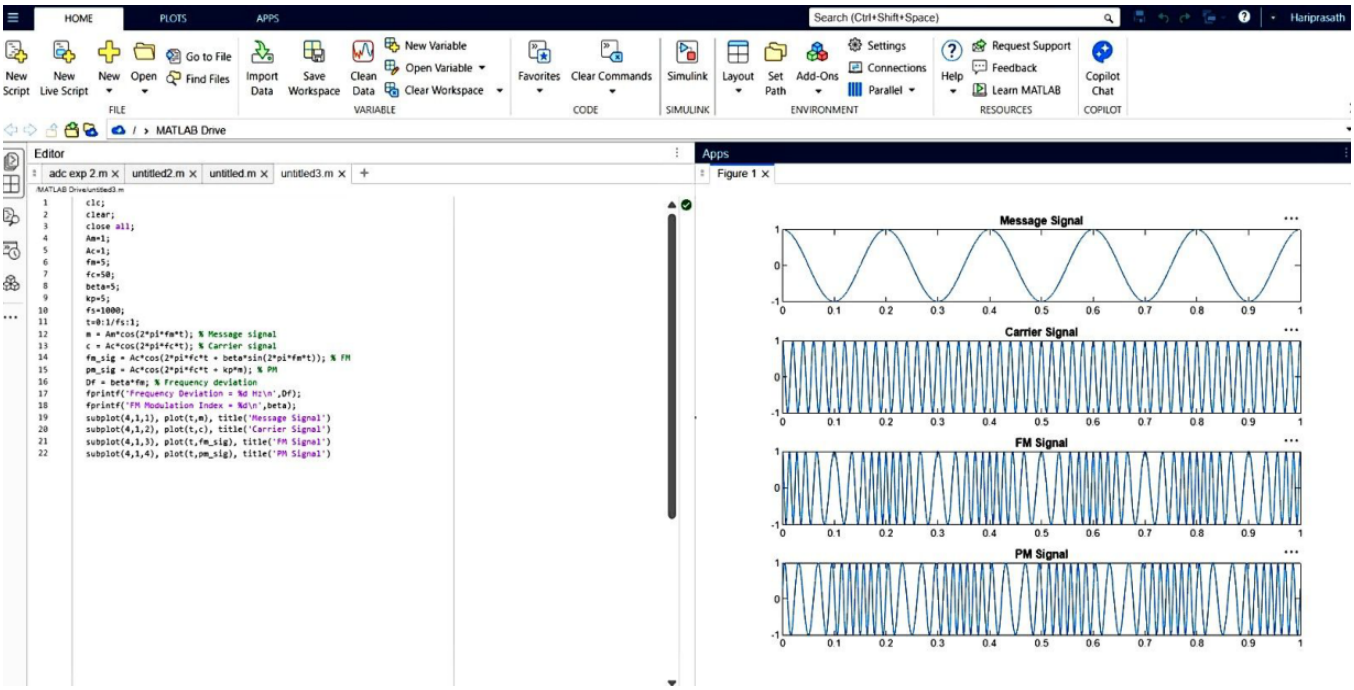
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
subplot(4,1,4), plot(t,pm_sig), title('PM Signal')
```

PROCEDURE

1. Open MATLAB and create a new script.
2. Enter the program code and run the script.
3. Observe the message signal and carrier signal.
4. Observe the FM and PM signals in the four subplots.
5. Note the frequency deviation and FM modulation index in the Command Window.

OUTPUT

MATLAB screenshot supplied by the student:



RESULT

The message, carrier, FM, and PM signals were generated and displayed in MATLAB.